Speech Emotion Recognition

Samuele Reale
Polytechnic University of Turin
s349519@studenti.polito.it

Lucio Fuoco
Polytechnic University of Turin
s348016@studenti.polito.it

Abstract

*This study presents a robust Speech Emotion Recognition (SER) system designed to handle class imbalance and acoustic variability. We propose a hybrid deep learning architecture, **CNN-BiLSTM-Attn**, which integrates Convolutional Neural Networks (CNN) for spatial feature extraction from Mel-spectrograms, Bidirectional Long Short-Term Memory (BiLSTM) units for temporal modeling, and a custom Attention mechanism to prioritize emotionally salient segments.*

Using the RAVDESS dataset, we implemented a speaker-independent evaluation protocol to ensure generalization across unseen subjects. To address the inherent scarcity of emotional data, we expanded the training set using offline pitch shifting and time stretching, while incorporating an online SpecAugment pipeline for frequency and time masking during the training process. Our results demonstrate that while baseline models on pure data suffer from severe overfitting with a 23.5% generalization gap, the integration of data augmentation effectively regularizes the network, nearly eliminating the gap to 0.26% in randomized splits. In rigorous speaker-independent tests, the model achieved a 64.58% validation accuracy, demonstrating high reliability in detecting universal high-arousal states such as Angry and Surprised. This study highlights the essential role of spectral regularization in mitigating actor-specific bias, providing a scalable framework for real-world, speaker-agnostic SER applications.

1. Introduction

Speech Emotion Recognition (SER) remains a pivotal challenge in human-computer interaction, requiring the extraction of nuanced emotional states from complex acoustic cues. While modern approaches often convert 1D waveforms into 2D **Log-Mel Spectrograms** to leverage the pattern recognition power of Convolutional Neural Networks (CNNs) [4, 8], treating spectrograms as static images ignores the inherently dynamic nature of prosody. Emotions such as the sudden energy intensification in anger or the gradual pitch decay in sadness are temporal phenomena that require sequential modeling.

To address these limitations, we propose a hybrid **CNN-BiLSTM-Attn** architecture. In this framework, CNN layers function as high-level feature extractors, identifying local spectral patterns which are then processed by Bidirectional Long Short-Term Memory (BiLSTM) units. This allows the system to capture temporal dependencies across both past and future contexts within an utterance. Central to our architecture is a **Custom Attention Mechanism** designed to overcome the limitations of standard recurrent models, which often assign equal weight to non-informative or silent segments. By dynamically weighting emotionally salient moments, the model mimics human auditory perception, focusing on the specific vocal inflections that carry the highest emotional entropy.

A significant hurdle in SER is the tendency of deep models to overfit to "actor fingerprints", memorizing specific vocal identities rather than universal emotional patterns. To ensure true generalization, this study employs a rigorous **Speaker-Independent (Actor-level) protocol** on the RAVDESS dataset [7]. We further mitigate this risk through a **Combinatorial Data Augmentation** strategy. By utilizing offline transformations (pitch shifting, time stretching, noise addition) to expand the training distribution and online **SpecAugment** masking to regularize the spectral input, we force the network to prioritize invariant acoustic features. This comprehensive approach addresses both class imbalance and speaker dependency, providing a robust framework for scalable, real-world SER applications.

2. Dataset

The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [7] is a widely recognized benchmark for Speech Emotion Recognition (SER) tasks. It is composed of high-quality recordings featuring 24 professional actors (12 male, 12 female), each performing two standard statements with a neutral North American accent.

For the scope of this study, we focus on the speech recordings, which encompass eight distinct emotional states: neutral, calm, happy, sad, angry, fearful, surprise, and disgust. Each emotion is provided at two levels of emotional intensity (normal and strong), with the exception of the neutral state. The controlled environment of the recordings ensures high acoustic clarity, making it a canonical choice for evaluating the effectiveness of deep learning architectures in recognizing prosodic patterns.

2.1. Motivation for Dataset Selection

The selection of RAVDESS for this study is driven by three critical factors. First, its **Lexical Control**: by restricting utterances to two semantically neutral statements, the dataset eliminates "lexical bias," forcing the model to rely exclusively on non-linguistic cues (pitch, timbre, rhythm) rather than semantic shortcuts. Second, the inclusion of **Emotional Intensity Levels** (Normal vs. Strong) provides a rigorous test for the model's sensitivity to subtle acoustic variations, distinguishing it from simpler datasets that only contain high-energy caricatures of emotion. Finally, the balanced gender distribution ensures that the learned features are robust across different fundamental frequencies (F_0), preventing gender-specific overfitting.

2.2. Data Pre-processing

To prepare the raw audio for the hybrid architecture, we applied a standardized pre-processing pipeline. First, silent intervals were removed using a trimming function with a threshold of 20 dB. To ensure a uniform input shape for the CNN layers, all utterances were standardized to a fixed duration of 3.0 seconds through zero-padding or central cropping.

The processed waveforms were then transformed into Mel-spectrograms using a Fast Fourier Transform (FFT) window of 2048 samples and a hop length of 512. We utilized 128 Mel bands to capture the spectral envelopes. The resulting power spectrograms were converted to decibel units (Log-Mel scale) and resized to a final dimensionality of $128 \times 128 \times 1$. This representation effectively compresses the acoustic signal into a feature map that highlights both harmonic structures and temporal energy shifts.

2.3. Data Augmentation

Given the limited size of the RAVDESS corpus, a randomized augmentation pipeline was implemented to enhance the generalization of the model. During training, each audio sample underwent a stochastic combination of the following transformations:

- **Pitch Shifting**: Altering the pitch by ± 2.5 semitones without affecting duration.
- **Time Stretching**: Modifying the playback speed within a range of 0.8x to 1.2x.
- **Gaussian Noise**: Adding random noise (factor 0.005–0.02) to simulate variable recording conditions.

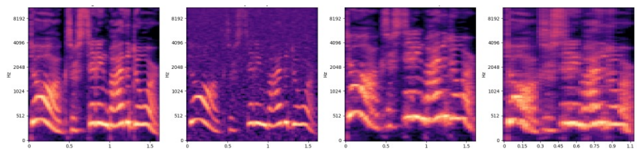


Figure 1: Visual comparison of Mel-spectrograms:from left to right the original signal is compared against 3 data augmentation strategies (Noisy, Pitch Shifted, Time Stretched).

- **Time Shifting**: Rolling the signal to prevent the model from overfitting to the starting position of the utterance.

To address the class imbalance, specifically the under-representation of the "Neutral" state, we applied a strategic oversampling. While standard emotional classes were augmented 7 times, the neutral class was sampled 13 times. The visual impact of these transformations on the acoustic features is illustrated in Figure 1, where the original Mel-spectrogram is compared with its augmented versions.

2.4. Dataset Configuration Parameters

To evaluate the impact of each optimization strategy, we organized the experimental phase into three distinct dataset configurations:

- **Baseline Configuration**: This setup utilizes the original RAVDESS utterances with a standard random 80/20 split.
- **Augmented Configuration**: We incorporated the combinatorial augmentation pipeline. The training set was expanded to approximately 9,000 samples, while the test set remained in its original form.
- **Speaker-Independent Configuration**: This represents the most challenging configuration, in which the dataset was split at the actor level following an 80/20 protocol (20 actors for training and 4 for testing).

The final Mel-spectrograms for all configurations were normalized to a range of $[0, 1]$ to ensure numerical stability during the training of the Attn-CNN-BiLSTM network.

3. Methodology

3.1. Model Architecture

To address the challenge of Speech Emotion Recognition (SER), we proposed a *Hybrid Attention-Based Convolutional Recurrent Neural Network (CRNN)*. While recent literature, such as Bhanbhro et al. [3], advocates for deep architectures utilizing up to six time-distributed convolutional blocks, our research focuses on a more streamlined, computationally efficient design. We hypothesize that a lighter architecture with aggressive regularization is better suited for limited datasets, minimizing the risk of overfitting while retaining the ability to capture complex spectro-temporal patterns.

Our architecture consists of three distinct stages: (1) **Spatial Feature Extraction** via a 2D CNN backbone, (2) **Temporal Modeling** using a Bidirectional LSTM, and (3) a **Learnable Attention Mechanism** to prioritize emotionally salient speech segments.

3.1.1 Spatial Feature Extraction (CNN Backbone)

The input to the model is a Log-Mel spectrogram of dimension $(128, 128, 1)$, representing frequency bins, time steps, and channels respectively. We employ a hierarchical Convolutional Neural Network (CNN) to extract high-level spectral features. Unlike deeper ResNet-style architectures, our encoder consists of three sequential convolutional blocks designed to reduce spatial dimensionality while increasing feature depth:

1. **Convolutional Layers:** We utilize 3×3 kernels with 'same' padding across all blocks to preserve spatial resolution during feature extraction. The number of filters increases geometrically ($32 \rightarrow 64 \rightarrow 128$) to capture increasingly abstract spectral correlations.
2. **Batch Normalization & Activation:** Each convolution is immediately followed by Batch Normalization to stabilize gradient flow, and a Rectified Linear Unit (ReLU) activation function ($f(x) = \max(0, x)$) to introduce non-linearity.
3. **Pooling & Regularization:** To reduce computational cost and introduce translation invariance, we apply Max Pooling with a 2×2 window. To combat overfitting a pervasive issue in SER tasks we inject Dropout at increasing rates (0.2 in early blocks, 0.3 in later blocks).

3.1.2 Feature Permutation and Reshaping

A critical step in our methodology is the transition from 2D image processing to 1D sequence modeling. The output of the final CNN block is a 4D tensor. To prepare this for the LSTM, we permute the dimensions to prioritize the time axis and collapse the frequency and filter dimensions.

If the output of the CNN is X_{cnn} with dimensions (B, F', T', C), where B is batch size, F' is frequency, T' is time, and C is channels, we reshape this to $(B, T', F' \times C)$. This allows the Recurrent Neural Network to treat the modified spectral features as a time-series sequence.

3.1.3 Temporal Modeling (Bi-LSTM)

Speech is inherently sequential, and emotional cues often evolve over time. To capture these long-term dependencies, we utilize a Bidirectional Long Short-Term Memory (Bi-LSTM) layer with 128 units. Unlike standard LSTMs, the bidirectional variant processes the sequence in both forward and reverse directions, allowing the network to utilize both past and future context for every time step:

$$h_t = [\overrightarrow{LSTM}(x_t);\ \overleftarrow{LSTM}(x_t)] \tag{1}$$

This results in a sequence of hidden states $H = \{h_1, h_2, ..., h_T\}$ that encapsulate the spectro-temporal context of the utterance, where $T = 16$ after convolutional downsampling.

3.1.4 Attention Mechanism

Emotions in speech are often characterized by short, high-intensity bursts (e.g., a sudden pitch rise) rather than uniform distribution. Standard RNNs often struggle to prioritize these segments. To address this, we implemented a custom **Self-Attention Layer** inspired by [2].

This layer learns a weighting vector α that assigns an "importance score" to each time step. The context vector c is computed as the weighted sum of the LSTM outputs:

$$e_t = \tanh(W_a h_t + b_a) \tag{2}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)} \tag{3}$$

$$c = \sum_{t=1}^{T} \alpha_t h_t \tag{4}$$

where W_a and b_a are learnable parameters. In our implementation, the attention bias b_a is defined per time step, allowing the model to learn positional importance across the temporal sequence. This mechanism allows the model to "focus" on the most emotionally relevant frames while ignoring silence or background noise.

3.1.5 Classification & Optimization

The attention-weighted context vector c is passed through a fully connected dense layer (64 units, ReLU) with high dropout (0.4) to further regularize the decision boundary. The final classification is performed by a Softmax layer over the $N = 8$ emotion classes.

3.2. Training Methodology

To ensure fair comparison and reproducibility, a standardized training protocol was established for all experimental configurations. The models were implemented using TensorFlow and Keras, utilizing the `tf.data` API to build efficient, asynchronous input pipelines.

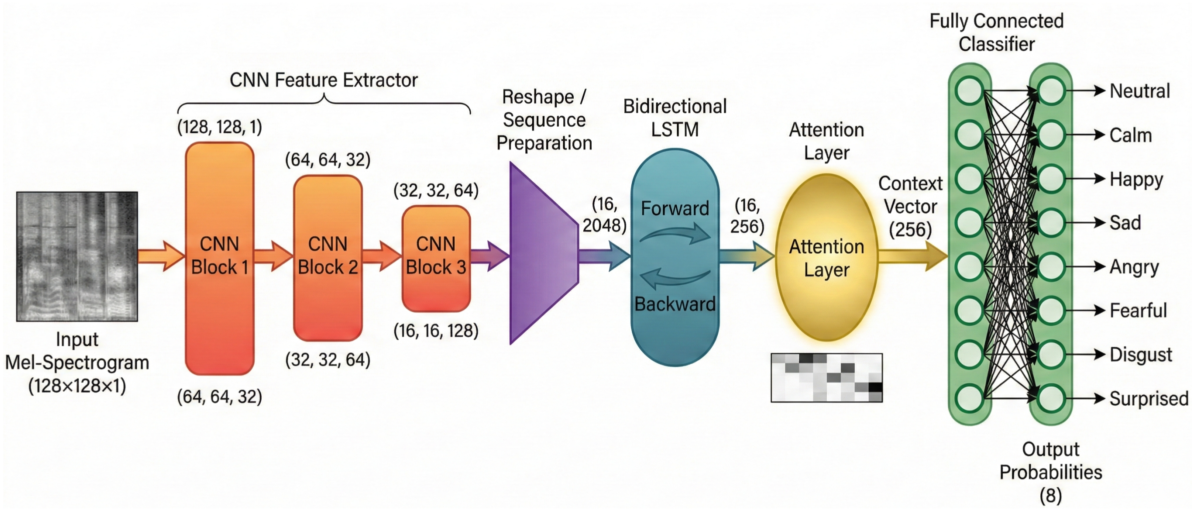


Figure 2: Overview of the proposed **CNN-BiLSTM-Attn** architecture.

Table 1: Layer Configuration of the Proposed Hybrid Model

Layer Type	Output Shape	Parameters
Input (Spectrogram)	$(B, 128, 128, 1)$	-
Conv2D Block 1	$(B, 64, 64, 32)$	3×3, ReLU, MaxPool
Conv2D Block 2	$(B, 32, 32, 64)$	3×3, ReLU, MaxPool
Conv2D Block 3	$(B, 16, 16, 128)$	3×3, ReLU, MaxPool
Reshape & Permute	$(B, 16, 2048)$	Flatten Freq & Channels
Bi-Directional LSTM	$(B, 16, 256)$	128 units, return seq
Attention Layer	$(B, 256)$	Custom Weighted Sum
Dense Layer	$(B, 64)$	ReLU, Dropout 0.4
Output Layer	$(B, 8)$	Softmax

3.2.1 Optimization & Hyperparameters

The training process was governed by the **Adam** optimizer [6] ($\beta_1 = 0.9, \beta_2 = 0.999$), selected for its ability to handle the sparse gradients inherent in high-dimensional spectrogram data.

Learning Rate Selection: We employed a fixed initial learning rate of $\eta = 0.001$. This value was determined following a preliminary coarse-to-fine search where we evaluated rates of 0.01, 0.001, and 0.0001.

- *High Learning Rate ($\eta = 0.01$):* Experiments with this rate resulted in unstable convergence. The validation loss exhibited significant oscillation, suggesting the optimizer was overshooting the global minimum due to excessive step sizes.

- *Low Learning Rate ($\eta = 0.0001$):* While stable, this rate led to prohibitively slow convergence and frequent stagnation in suboptimal local minima, failing to reach competitive accuracy within the epoch limit.

- *Optimal Rate ($\eta = 0.001$):* This setting provided the ideal balance, facilitating rapid initial descent down the error surface while maintaining stability during fine-tuning.

Objective Function: To optimize the network parameters, we employed the **Categorical Cross-Entropy (CCE)** loss function. This choice was motivated by the probabilistic nature of the Softmax output layer, where the goal is to maximize the likelihood of the correct emotion class y_c given the input spectrogram.

Mathematically, CCE is defined as:

$$\mathcal{L}_{CCE} = -\sum_{i=1}^{N} y_i \log(p_i) \tag{5}$$

where $N = 8$ represents the emotion classes, y_i is the binary indicator (0 or 1), and p_i is the predicted probability.

3.2.2 Adaptive Learning Strategy

To prevent overfitting, a critical risk given the relatively small size of the RAVDESS dataset, we implemented a strict regularization policy using dynamic callbacks. These callbacks monitor the validation loss ($\mathcal{L}_{val}$) and accuracy (Acc_{val}) at the end of every epoch:

- **Model Checkpoint:** The model weights are saved only when Acc_{val} improves, ensuring that the final evaluated model represents the peak generalization performance rather than the state at the final epoch.

- **ReduceLROnPlateau:** If $\mathcal{L}_{val}$ fails to improve for 5 consecutive epochs, the learning rate is reduced by a factor of 0.5 (e.g., $0.001 \to 0.0005$). This allows the optimizer to make finer adjustments when settling into a local minimum.

- **Early Stopping:** Training is automatically terminated if $\mathcal{L}_{val}$ does not improve for 12 consecutive epochs. This prevents the "memorization" phase where training loss continues to drop while validation loss diverges. The maximum number of epochs was set to 60.

3.2.3 Online SpecAugment

For the augmented experimental configurations, we implemented an *Online SpecAugment* pipeline. Unlike traditional offline augmentation (which increases dataset size on disk), this technique is applied on-the-fly during batch generation. We utilized two specific masking policies:

1. **Frequency Masking:** A block of consecutive frequency channels $[f_0, f_0 + f)$ is masked (set to zero). The width of the mask f is sampled from a uniform distribution $U(0, F)$, where the hyperparameter $F = 20$ defines the maximum bandwidth to be removed (approx. 15% of the frequency spectrum).
2. **Time Masking:** A block of time steps $[t_0, t_0 + t)$ is masked. The duration t is sampled from $U(0, T)$, where $T = 20$ defines the maximum temporal duration to be masked. This forces the network to learn robust features that do not rely on contiguous speech segments.

By applying these masks dynamically, the model rarely encounters the exact same feature map twice, effectively regularizing the network against noise and improving robustness across different speakers.

4. Experiments and Results

4.1. Experimental Setup

To rigorously evaluate the proposed architecture and validate the efficacy of our augmentation strategies, we designed three distinct experimental scenarios. These experiments link the dataset configurations defined in Section 2 with the training protocols established in Section 3.

4.1.1 Scenario Descriptions

- **Experiment I (Baseline):** This scenario serves as the control group. The model is trained on the *Baseline Configuration* (pure RAVDESS utterances) using a standard stratified 80/20 split. No online SpecAugment is applied. This establishes the fundamental capacity of the hybrid architecture to learn from limited, clean data.
- **Experiment II (Data Diversity):** This scenario introduces the full augmentation pipeline. The model is trained on the *Augmented Configuration* (9,000 samples) and subjected to Online SpecAugment (freq/time masking) during training. The goal is to quantify the performance gain attributed purely to increased data diversity and regularization.
- **Experiment III (Generalization):** This is the most rigorous evaluation. Using the *Speaker-Independent Configuration*, the model is trained on Actors 1–20 and tested strictly on Actors 21–24. This simulates a real-world deployment where the system must recognize emotions from users it has never encountered during training.

4.2. Results and Discussion

To provide a comprehensive evaluation of the model's performance, we report the **Weighted Average** of Precision, Recall, and F1-score. Unlike the macro average, which treats all classes equally, the weighted average accounts for *support*, the number of true instances for each emotional label. By weighting each class's score by its frequency in the test set, these metrics provide a more accurate representation of the system's real-world reliability across the entire emotional spectrum of the RAVDESS dataset.

4.2.1 Baseline and Augmentation Analysis (Exps I & II)

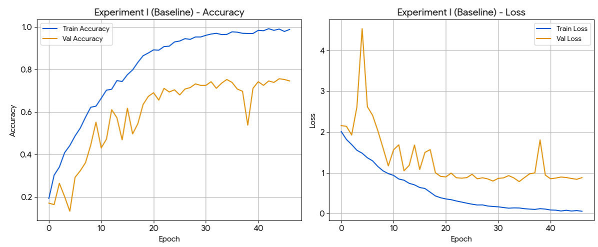
Experiment I established a performance benchmark using raw audio. While the model reached a peak training accuracy of 99.26%, the validation accuracy stalled at 75.69%, resulting in a 23.5% generalization gap (see Fig. 3a). This significant disparity suggests that the model "memorized" training samples by recognizing specific "actor fingerprints" rather than learning generalized emotional prosody. Classes like **Calm** (F1: 0.85) and **Angry** (F1: 0.81) performed well due to distinct energy pro-

Table 2: Overall Performance Across Experimental Scenarios

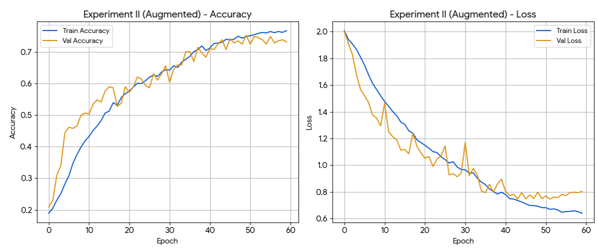
Experiment	Train Acc	Val Acc	Gap	F1-Score*	Precision*	Recall*
Exp I (Baseline)	99.26%	75.69%	+23.57%	0.76	0.77	0.76
Exp II (Augmented)	75.09%	75.35%	-0.26%	0.75	0.76	0.75
Exp III (Spk-Indep.)	73.08%	64.58%	+8.50%	0.64	0.68	0.65

*Metrics are reported as the **weighted average** based on class support.

files, whereas **Sad** (F1: 0.65) suffered due to acoustic overlap with *Fearful* and *Calm*.



(a) Exp I: Severe overfitting on pure data.



(b) Exp II: Regularization via augmentation.

Figure 3: Comparison of training histories for Experiments I and II, showing the transition from divergent to convergent accuracy and loss curves.

In Experiment II, the introduction of a 9,000-sample augmented dataset and the SpecAugment pipeline effectively regularized the model. The generalization gap was nearly eliminated, dropping to a mere 0.26% (75.09% training vs. 75.35% validation accuracy), as illustrated in Fig. 3b. Although the training accuracy was intentionally reduced to mitigate pixel-level memorization, the high validation performance still partially relied on the recognition of actor-specific timbre. Because the dataset utilized a randomized split, the model encountered the same vocal identities in both training and validation sets, allowing it to leverage familiar speaker characteristics alongside the generalized emotional patterns forced by the augmentation.

4.2.2 Speaker-Independent Generalization (Exp III)

Experiment III represents the core contribution of this work, testing the model on actors entirely excluded from the training set. The model achieved a validation accuracy of 64.58%. While the generalization gap rose to 8.5%, it remained significantly narrower than the baseline, indicating a successful transition to learning speaker-invariant features.

The analysis of emotional labels, as detailed in Table 3, reveals a clear hierarchy in universality:

- **Universal High-Arousal Cues:** The model excelled in identifying **Surprised** (F1: 0.86, Recall: 0.94), **Disgust** (F1: 0.83), and **Angry** (F1: 0.77). These results suggest that high-arousal, aggressive, or sharp vocal signatures are highly transferable across different individuals.

- **Speaker-Dependent Volatility:** Subtle emotions like **Sad** (F1: 0.42) and **Fearful** (F1: 0.47) experienced significant degradation. The low recall for **Fearful** (0.31) suggests that "shaky" vocal qualities are interpreted inconsistently across different actors.

- **Arousal Baselines: Calm** retained a respectable F1-score of 0.72, yet **Neutral** remained difficult (F1: 0.51). This suggests that the acoustic default, the pitch and energy floor of a speaker, varies so significantly between individuals that the model struggles to define a universal "Neutral" category. This leads to frequent confusion between low-arousal states when the system encounters an unfamiliar vocal range, as clearly visible in Fig. 4.

Table 3: Detailed Classification Metrics for Speaker-Independent Evaluation (Exp III)

Emotion	Precision	Recall	F1-Score	Support
Neutral	0.43	0.62	0.51	16
Calm	0.63	0.84	0.72	32
Happy	0.46	0.50	0.48	32
Sad	0.41	0.44	0.42	32
Angry	0.80	0.75	0.77	32
Fearful	0.91	0.31	0.47	32
Disgust	0.92	0.75	0.83	32
Surprised	0.79	0.94	0.86	32
Weighted Avg	**0.68**	**0.65**	**0.64**	**240**

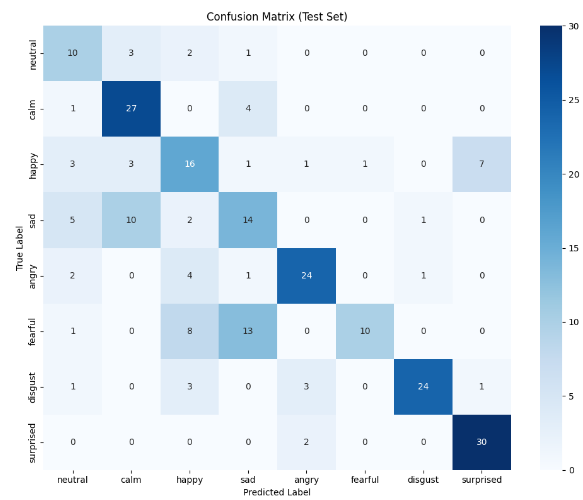


Figure 4: Confusion Matrix for the Speaker-Independent evaluation (Exp III).

4.2.3 Research Implications

While a 64.58% speaker-independent accuracy indicates that further optimization is necessary for mission-critical use, it provides a strong proof-of-concept for real-world SER. The model's reliability in identifying high-arousal states like *Anger* and *Surprise* suggests immediate utility for preliminary sentiment filtering in environments such as customer service or mental health monitoring. By successfully mitigating actor-specific bias, this hybrid architecture represents a significant step toward speaker-agnostic systems, though challenges remain for reliable, large-scale deployment.

5. Conclusion and Future Work

This study aimed to develop a robust Speech Emotion Recognition (SER) system capable of generalizing beyond the specific vocal characteristics of the training subjects. By evaluating a hybrid **CNN-BiLSTM-Attn** architecture under three distinct experimental protocols, we have isolated the factors that contribute to the "generalization gap" in deep emotional modeling.

5.1. Key Findings and Insights

The transition from Experiment I to Experiment II provided a critical insight: overfitting in SER is largely driven by spectral memorization. Although the baseline model achieved near-perfect training accuracy (99.26%) by memorizing actor-specific timbres, it did not generalize. The introduction of our Combinatorial Data Augmentation pipeline effectively regularized the network, reducing the generalization gap to just 0.26%. However, Experiment III revealed the inherent limitations of supervised learning on small datasets. The drop to 64.58% accuracy on unseen speakers highlights a clear hierarchy of universality:

- **Universal Anchors:** High-arousal states like *Anger* and *Surprise* possess robust, speaker-invariant acoustic signatures (sharp onsets, high energy) that are easily transferred across individuals.

- **The Calibration Problem:** Low-arousal states (*Neutral*, *Calm*, *Sad*) rely heavily on an individual's "acoustic zero-point." Without a reference baseline for a new speaker's natural voice, the model struggles to distinguish a "sad" drop in pitch from a speaker who simply has a naturally deep voice.

5.2. Challenges and Future Directions

The primary challenge identified is the Precision-Recall trade-off in subtle emotions. While our attention mechanism successfully isolated salient emotional segments, it could not fully compensate for the lack of speaker-specific context, leading to the low recall in *Fearful* (31%) and *Sad* (44%).

To bridge this final gap, future research should move beyond static supervised learning. We propose two key extensions:

1. **Self-Supervised Learning (SSL):** Integrating pre-trained embeddings from large-scale models like *Wav2Vec 2.0* [1] or *HuBERT* [5]. These models, trained on thousands of hours of unlabeled speech, could provide the rich, contextual representations necessary to disentangle speaker identity from emotional content.

2. **Few-Shot Speaker Adaptation:** Implementing a calibration phase where the model receives a few seconds of a user's "Neutral" speech before inference. This would establish a dynamic baseline, potentially resolving the confusion between *Calm*, *Neutral*, and *Sad*.

Ultimately, this study demonstrates that while augmentation can bridge the *statistical* gap between training and validation, solving the *semantic* ambiguity of subtle emotions requires models that do not merely process speech, but actively adapt to the speaker behind it.

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